

MTH212M PRACTICE PROBLEMS WEEK 2

Question 1. Fix $p \in (0, 1)$ and let $\{X_n\}_{n \geq 1}$ be an i.i.d. sequence of *Bernoulli*(p) RVs. Define $Y_n := \min\{10, X_1 + X_2 + \cdots + X_n\}, n = 1, 2, \dots$. Identify the communication classes for the Markov Chain $\{Y_n\}_{n \geq 1}$ and compute the period of states.

Question 2. Recall the two person's Zero-sum game discussed in the class. Assume that initially the two persons A and B start with 10 Rupees and 5 Rupees respectively. Let X_n denote the amount of money held by A after the n -th toss. If a fair coin is used in the tosses, then find the following:

- (i) the law of X_3
- (ii) $\mathbb{P}(X_1 = 9 | X_3 = 7)$
- (iii) communication classes
- (iv) period of states

Question 3. A urn contains b black balls and w white balls. Draw a ball at random, and replace it by a ball of the other colour. Let X_n denote the number of black balls after the n -th draw. Identify the communication classes for the Markov Chain $\{X_n\}_{n \geq 1}$ and compute the period of states.

Question 4. Suppose $\{X_n\}_{n \geq 0}$ denotes the symmetric 1-dimensional Random walk on the set of integers. Identify the communication classes and compute the period of states.